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Optimized evolutionary algorithm and supervised ACO mechanism to mitigate attacks and improve performance of adhoc network

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ABSTRACT

Optimization is the procedure of discovering optimal solution or results under particular circumstances. Typically, optimization is utilized for maximization or minimization of values of functions; it can be local or global optima. Discovering exact solutions to the issues is NP-hard. This type of complicated issue needs exponential quantities of computation power as well as item and quantity of decision parameters rise. For overcoming the issues, research scholars have suggested Evolutionary Algorithm (EA) protocols as a way for searching near-optimal solutions. In this work, Ant Colony Optimization (ACO) is used combined with the power of faith on effective factors in distance of trustworthy optimum routing in WSN. ACO has its basis in the foraging behavior of ants who look for the shortest route between then nest and the food source. The energy efficient node random trust based on routing network is designed in a safe and random routing mode that reveals all fingerprints and useful load distribution. The details are combined with confidence choices, using a modified ACO algorithm. The ACO application, makes this method a stronger and faster with a metaheuristic algorithm compared with the other routing methodologies. Modified ACO method, exercise as a parameter for calculating the energy levels of terminals used fitness route planning its organization. Proposed work shows improvement in the network performance under Dos and DDoS attack in terms of amount of data loss, data delay.

1. Introduction

Development of routing algorithms for MANET is a problematic task due to several reasons. Various proactive as well as reactive algorithms are suggested for fulfilling several goals such as effective usage of bandwidth, rapid path convergence, freedom from loops and so on. The issue of energy effective routing for improving network life time is an ongoing one. Energy resources of active nodes get drained faster than others which might result in network partitioning, therein leading to decrease in network life time. Three primary methods of improving node lifetime are (1) effective management charge, (2) data power transmission management, and (3) system design data or power management [1].

Energy effective design of algorithms is created through differing the transmission range of nodes. Varying transmission range implies the controlling of power levels of all packets in a distributed manner in all nodes, thereby impacting energy utilization of networks [2]. Selecting greater transmission range decreases quantity of nodes required to reach destination however generates huge interference, while decreasing transmission range needs greater quantity of forwarding nodes, resulting in lesser energy usage [3].

Ant colony optimization technique is an evolutionary algorithm helps to find the perfect path for the data transmission. Since it helps to find the exact path with higher power transmission and less distance between the intermediate nodes, it ensures successful delivery of data without unnecessary overhead and delay. Hence ant colony optimization technique helps to find the best and shortest path.

Optimization is the procedure of discovering optimal solution or results under particular circumstances. Typically, optimization is utilized for maximization or minimization of values of functions; it can be local or global optima. In optimization, there are various kinds of issues such as linear, non-linear, dynamic and so on and all have their own way of solution [4].

2. Literature survey

An Optimization is the process where additional features are added to the existing system to make it function better than the earlier. Optimization can be done in process, algorithms, individual system, and network to improve the performance of the existing one. Optimizing

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systems as well as procedures is significant for efficacy as well as economics of science and engineering domain. Optimization issues are resolved through the usage of rigorous or approximate mathematical search methods. The former utilized linear, integer, dynamic or branch-and-bound methods for arriving at optimal solution for moderately sized problems. But, optimization of real life issues of the scale typically faced in engineering is more problematic due to complicated as well as large solution spaces. Discovering exact solutions to the issues is NP-hard [5]. This type of complicated issue needs exponential quantities of computation power as well as item and quantity of decision parameters rises. For overcoming the issues, research scholars suggested EA protocols as a way for searching near-optimal solutions.

EAs are stochastic search techniques which simulate the behavior of biological species which display optimized behavior guided by learning, adaptation as well as evolution [6]. Computation systems which simulate activity of ants, bees, frogs and so on are utilized as a way to rapidly search for more optimal solutions to complicated optimization issues.

The computer design with intelligence question with optimization problem to computational sub-domain used of Swarm Intelligence (SI) there are easy for answer. SI primarily focuses on PSOA, ACO and ABC. SI-based methods are biologically inspired. Swarms are sets of mobile agents which resolve issues together. SI is the characteristic of the system wherein group activity of individual agents leads to complicated global patterns.

All individuals in the swarm have simple rules of action as well as access to restricted quantity of data through immediate neighbors or local environment. As well food difficult in problem to accomplish swarm methods used for optimization. The paradigms simulate the activities of searching for food, way of living as well as style of defense to be employed in computation problems. Almost all real problems can draw analogies in SI based algorithms [7].

Because of its nature, infrastructure, topology as well as functionality of adhoc wireless networks, SI methods are appropriate for routing as well as energy resource optimization related problems in MANET as well as WSN [8]. Bio-inspired, SI methods are encouraging for adhoc as well as WSNs because of the reasons given below: (1) Locality of interactions, (2) Availability of several routes, (3) Self-organizational behavior, (4) Failure backup, (5) Capacity to adapt in a rapid as well as resilient manner to topological as well as traffic change and component failure, (6) Scalable performance resilience to failure, (7) Loss internal to the algorithm, (8) Easiness of design as well as tuning.

A Survey on Different Algorithms of Ant Colony Optimization used to propose a model to defend against DDoS attack [9]. Associative classification algorithm yields better understanding of ant colony optimization against network attacks [10]. Decision Support System supports to build a model to analyze normal traffic from abnormal one [11]. Implementation of efficient routing protocols for MANETs is a challenging task. Routing requirements such as higher packet rate, low data loss and adaptive network topology provides ensured network security [12]

The basic ACO metaheuristic discussed in the earlier work proves why these kinds of protocols function well in mobile multihop adhoc networks. They take into consideration several significant characteristics of MANET.

Dynamic topologies: This characteristic handles with ant performance multi-hop networks various design used for computer with identification foods. ACO metaheuristic has its basis in agent systems and works with single ants. This permits a higher adaptation to current network topology. **Local work**: as opposed to other routing methods, ACO metaheuristics have their basis in local information.

Link quality: it is possible to perform integration of link quality into calculation of the pheromone integration particularly into evaporation procedure which enhances decision process with regard to link quality.

Support for multipath: all nodes have routing tables with entries for every neighbors that comprise pheromone concentrations. Decision rules for choosing subsequent nodes have their basis in pheromone concentration on current node that is provided for all possible links. Hence, the method supports multipath routing.

Here, data packets are the artificial ants sent from sources as an optimal route to find the target node of each cycle.

3. Ant Colony Optimization (ACO)

The ant behavior most of generic problem initialize by this methods (ACO) metaheuristic. Ants from their nests in search of food to start out with is a situation where location [2]. In choosing ant static way point. In their way, but, the pheromone scent of other ants to follow the path that will allow the deposit of a chemical substance pheromone. Sometime ant's group arrearage in route static location selective, a link intense pheromone deposit, bypassing the nest, instead of the nest, the path of the ants allows the path to be more attractive. For the link evaporates pheromone after certain amount of time. There are long route selective ant in start location to end location on going ant has been lost fast than shortest path. To find food more ants [13].

One of the important factors to consider is that the pheromone level and time interval pheromone perfume that is evaporated at a particular time is strong. There are network alignment of mobile agents can continue to work as we collaborate to make ants jointly work. Every ant in network form in static or dynamic. Node or ants grouping desire or join ant's position in the network. This analogy is more robust and falls directly into any at hoc networks that are connected from point of view.

Between exploration and exploitation of any free data search stochastic for success greatly transmission node a harmonious data. But all of the meta-heuristic algorithms, these two issues are crucial to the conflicting. Search promising unexplored areas of space exploration in the search for solutions while expanding exploitation and effectively in the search for good solutions is to use. The ants on past exploits reinforcement the synthetic pheromone found a good solution [14]. However, to facilitate greater reinforcement of the focal premature. The following methods have been suggested to maintain the diversity of the search space.

(a) The ant control trail pheromone exponent.
(b) In random route in major processing.

A quantity of pheromone - $\Delta\tau_{i,j}$, fixed function depends on the pheromone path their route completing in node incremental path usage. (i, j) pheromone of amount, node moving i and j as follows [15]:

$$\tau_{i,j} = \sigma.\tau_{i,j} + \Delta\tau_{i,j} \tag{1}$$

where, σ – Evaporation pheromone factor. It lowest number 1 to pheromone accumulation avoid.

At i first point, which is a choosing point j (i.e. According to the probability of the path (i, j)) follows:

$$P_{i,j} = \frac{(\tau_{i,j})^{\alpha}.(\eta_{i,j})^{\beta}}{\sum_{(i,k)\in C}(\tau_{i,k})^{\alpha}.(\eta_{i,k})^{\beta}} \tag{2}$$

where,
$\tau_{i,j}$: Path of intensity pheromone,
$\eta_{i,j}$: (i, j) field of visibility ant's,
α and β: relative importance parameters which control pheromone intensity,

Set of point starting with possible paths i ((i, k) is a path of C).

The central characteristics are attributable to ACO events for routing problems in the following package:

- Multipath routing data transmission and adaptive traffic.
- The message reply active information.
- Node use stochastic components.

• Local values do not allow global impacts.

- Set by the pure shortest path operation with load balanced.
- To view the sensitivity.

3.1. Proposed ACO optimized trust based energy aware routing

The energy efficient node random trust based on routing network is designed in a safe and random routing mode that reveals all fingerprints and useful load distribution. ACO algorithm using this modified route selective are performance in incorporated with trust details. The main usage of this algorithm fast while comparing routing method for different result showing the proposed system ACO high level range based on robust method.

Once a broadcast request is not started, ACO starts at the core ends nodes. Which is a variant ant system this method uses the ant density algorithm. Ant system when the base nodes are select the next node is probabilistically on the distance and pheromone intensity. So it is possible to make sure that this method tries to produce dynamic and random routes.

Modified ACO method, the fitness calculated in the parameter for the incorporating the energy levels, is route identify the node. The initiation using ACO on-demand routing process with route transmission [16].

The network initial nodes according to estimated size. The uniform distribution function using ants distributed in the nodes. To recording its node moving network point, tabu list is maintained for each ant. The search node size of tabu list for fixed space. Ant that currently resides on the primary node is added to the list of Tabu. Exercise values of all unused terminals on the network are identified, and the ant movement is stimulating the fitness to the best competition. To n_1 movement of ant node for identify by calculated the probability values in n_1 fitness (see Fig. 1).

To consider number of ants (m) many node create network n then probability ant m_i in choosing a node for given by,

$$p_{ij}(t) = \frac{\left[\tau_{ij}(t)\right]^{\alpha} \cdot \left[\eta_{ij}\right]^{\beta} \cdot \left[\varepsilon_j\right]}{\sum_{j=1}^{n} \left[\tau_{ij}(t)\right]^{\alpha} \cdot \left[\eta_{ij}\right]^{\beta} \cdot \left[\varepsilon_j\right]} \tag{3}$$

where, τ_{ij} is the intensity pheromone in the node
ij ε_j is in the node charge contained
The range of node visibility η_{ij}.
Visibility respectively pheromone in given the values α and β
The fitness function is computed using Eq. (4):

$$f(x) = \left(\frac{e_i - \mu}{\sigma}\right)/N$$

where

e_i is the end to end delay for the selected route
μ is the average end to end delay across all routes
σ is the Standard Deviation in the delays across all routes
N is the number of hops to destination

(4)

In this work, a novel η value based on distance between nodes and delay is proposed and is given by

$$\frac{PDR}{t_{sd} \times p_{sd}}$$

where

t_{sd} is the sum of trust of all nodes distance between source to destination
p_{sd} is the sum of power consumed for a packet to reach from source to destination
PDR is the Packet Delivery Ratio

(5)

The ants moved from the neighbors and also to increase the probability of entry gets specific probability. Compare the quality and efficiency of neighbors' probability values to neighbors in the neighboring energy of living neighborly connections. The various nodes collect from the request node energy consumption information the ACO method collecting. That varies between true and false information about the node, expressed in terms of normal index. Ant edges collects information about the status of the request on the way and determines the overall trend of product and personal terminals reliability. When an ant is targeted, the target grades will protect the path quality against the value of reference. The answer is based on the quality obtained by ants, the intermediate edges refresh the pheromone values. The purposed system of this evaporation is to enable the main nodes or data complete node to identify neighbornode in each path sent by wireless sensor network [17].

Three optional variant of this approach utilizes ACO. So exploration and exploitation can be varied levels of regulations. This approach sets a threshold of 30% the levels of exploitation and 70% of study. So who wants to explore the means of exploitation. This node enables a random connecting mechanism to the selection session node. If the approach is enabled for exploitation, the procedure will enable it to choose the best bathroom that goes far and the review system is enabled, it find out the Cumulative Distribution Function (CDF) using this tries algorithm.

After the identity of the whole path, the path from each ant has been selected and the least route is identified as the path to the best path and the final route is used. The nodes in the final path are reduced to each energy level. To identify the values of fitness new energy levels used for next transmission. Therefore, if a node is used, its fitness value gradually reduces the probability of selection of terminal selection. This program leads to a node probability reduction, which leads to self-turning that uses a standard way of all nodes. This approach leads to a breakdown of the lower node, as long the network lifetime.

Table 1
Data loss rate in DOS for 0% to 90% of compromised nodes with proposed optimized trust technique.

Percentage of compromised nodes	Proposed trust energy aware routing	Optimized trust energy aware routing
0	0.0891	0.0816
10	0.098	0.0893
20	0.1072	0.0996
30	0.1131	0.1073
40	0.1374	0.1267
50	0.3405	0.3182
60	0.4633	0.4308
70	0.6065	0.5525
80	0.6836	0.6373
90	0.8127	0.7477

4. Results and Discussion

The simulations of compromised nodes (0%–90%) investigated for proposed system energy were conducted for varying percentages energy aware routing. For each simulation results showing for amount of data loss, data delay, detection probability or percentage of detected nodes in DOS & DDOS data drop and losing data are presented in Tables 1 to 6 and Figs. 2 to 7.

From Table 1 as well as Figs. 2a & 2b, the proposed energy aware increase by the data loss rate in DOS by 7.35% for 20 percent of compromised nodes, by 8.1% for 40 percent of compromised nodes, by 7.26% for 60 percent of compromised nodes, and by 7.01% for 80 percent of compromised nodes when compared with optimized trust energy aware routing.

From Fig. 3, it can be observed that the detection probability has higher value by various false positive probabilities.

From Fig. 4, it can be observed that the percentage of detected nodes has lower value by various percentages of compromised nodes.

From Table 4 as well as Figs. 5a & 5b, it can be observed that the proposed trust energy aware routing increased end to end in DOS by 6.01% for 20 percent of compromised nodes, by 8.38% for 40 percent of compromised nodes, by 8.73% for 60 percent of compromised nodes, and by 8.25% for 80 percent of compromised nodes when compared with optimized trust energy aware routing.

From Table 5 as well as Figs. 6a & 6b, the confidence that it has proposed is to note that the rate of loss of data on the routing data has

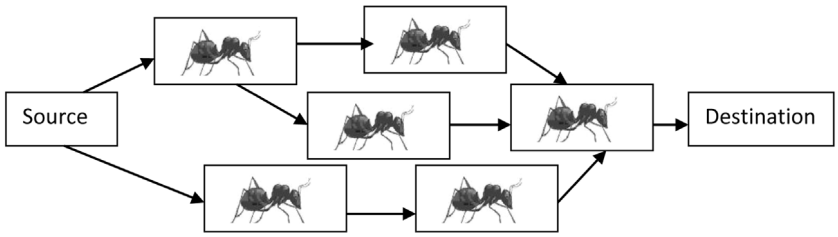


Fig. 1. Ant colony — considers trust, energy and shortest path.

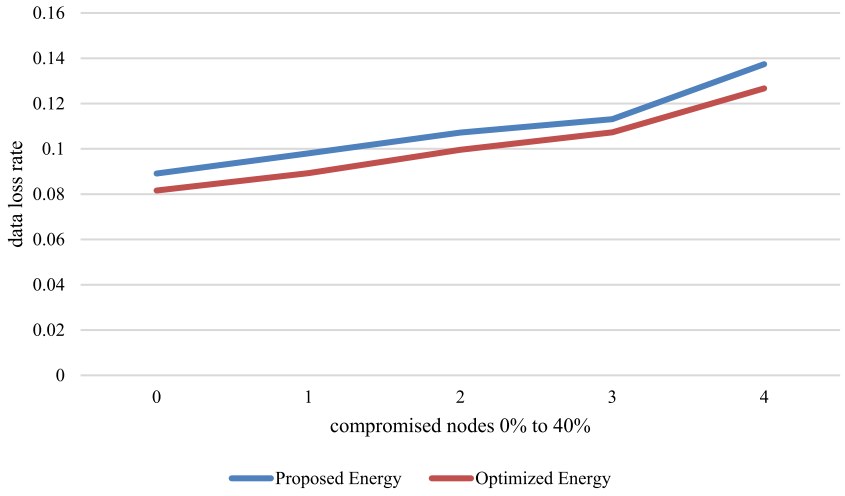


Fig. 2a. Data loss rate in DOS for 0% to 40% of compromised nodes with proposed optimized trust technique.

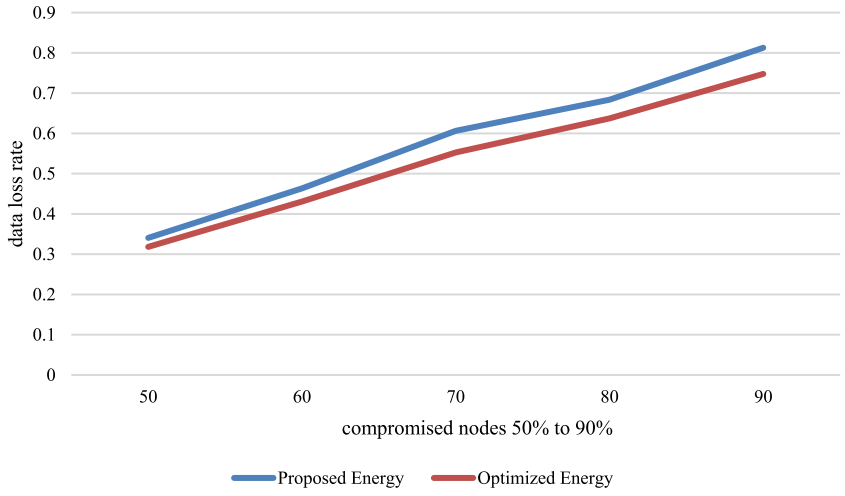


Fig. 2b. Data loss rate in DOS for 50% to 90% of compromised nodes with proposed optimized trust technique.

Table 2
Detection probability in DOS for 0% to 90% of compromised nodes with proposed optimized trust technique.

False positive probability	Detection probability
0	0.76
0.03	0.8
0.06	0.81
0.09	0.86
0.12	0.9
0.15	0.95
0.18	0.97
0.21	0.99

increased in DDOS by 9.39% for 20 percent of compromised nodes, by 5.64% for 40 percent of compromised nodes, by 6.61% for 60 percent of compromised nodes and by 9.25% for 80 percent of compromised nodes when compared with optimized trust energy aware routing.

From Table 6 as well as Figs. 7a & 7b, it can be observed that the proposed trust energy aware routing increased end-to-end in DDOS by 5.73% for 20 percent of compromised nodes, by 9.04% for 40 percent of compromised nodes, by 5.3% for 60 percent of compromised nodes and by 9.27% for 80 percent of compromised nodes when compared with optimized trust energy aware routing.

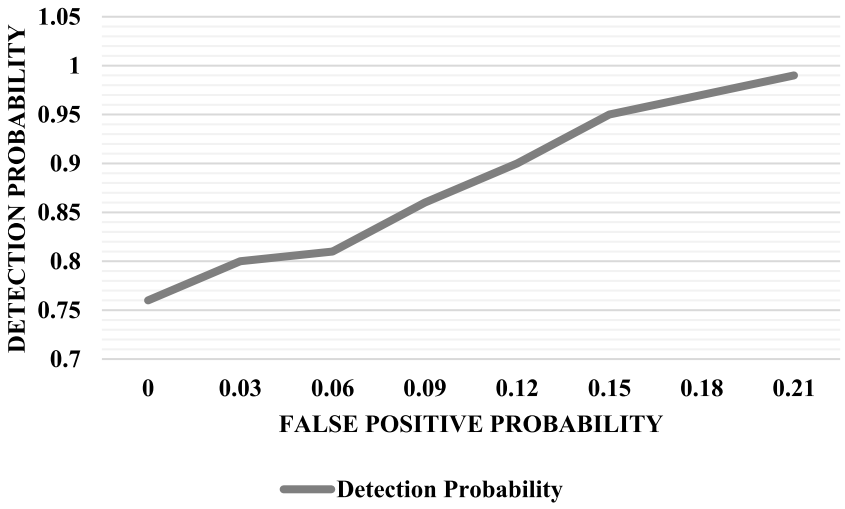


Fig. 3. Detection probability in DOS for 0% to 90% of compromised nodes with proposed optimized trust technique.

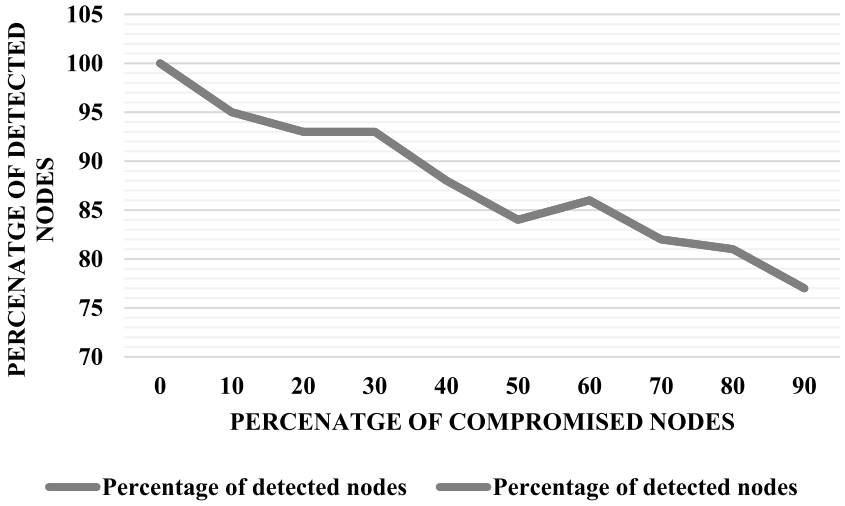


Fig. 4. Percentage of detected nodes in DOS for 0% to 90% of compromised nodes with proposed optimized trust technique.

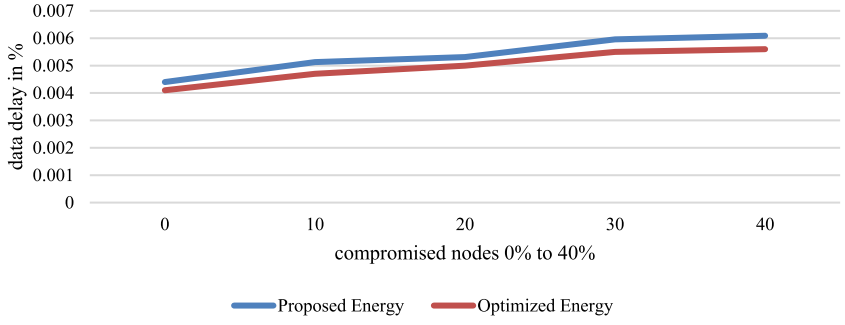


Fig. 5a. Data delay in DOS for 0% to 40% of compromised nodes with proposed optimized trust technique.

5. Summary of Result

In the work a trust based mechanism was proposed to improve the QoS. The proposed trust based routing improved the overall routing performance even when no malicious nodes were present showing its suitability for networks without malicious nodes. In the final phase of the work optimization was used to select the optimal parameters of trust using Ant Colony metaheuristic. The results showed that the proposed trust energy aware routing increased end to end in DOS by 6.01% for 20 percent of compromised nodes, by 8.38% for 40 percent of compromised nodes, by 8.73% for 60 percent of compromised nodes and by 8.25% for 80 percent of compromised nodes when compared with optimized trust energy aware routing and the proposed trust energy aware routing increased end to end in DDOS by 5.73% for 20 percent of compromised nodes, by 9.04% for 40 percent of compromised nodes, by 5.3% for 60 percent of compromised nodes, and by 9.27% for 80 percent of compromised nodes when compared with optimized trust energy aware routing. Similar improvements in end-to-end delay were also observed.

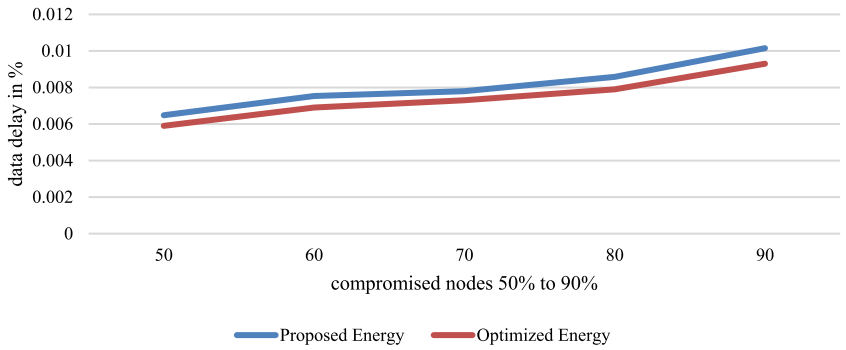


Fig. 5b. End to end delay in DOS for 50% to 90% of compromised nodes with proposed optimized trust technique.

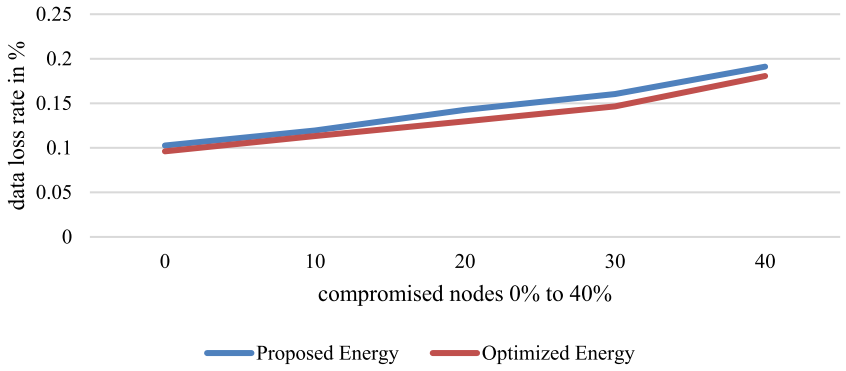


Fig. 6a. Data loss rate in DDOS for 0% to 40% of compromised nodes with proposed optimized trust technique.

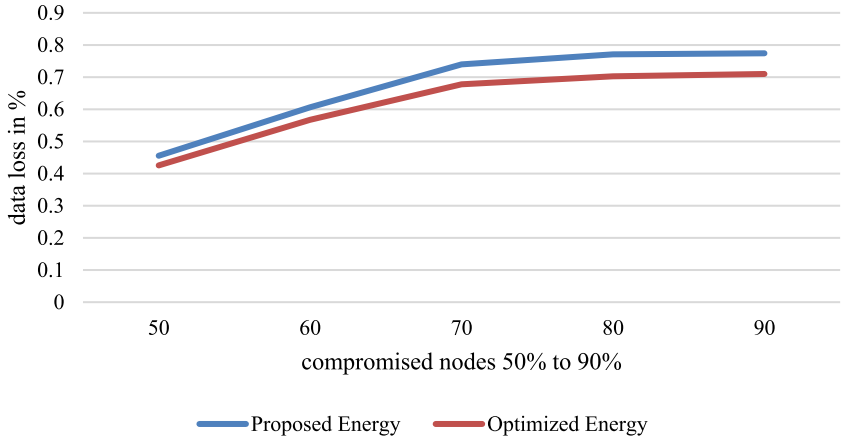


Fig. 6b. Data loss rate in DDOS for 50% to 90% of compromised nodes with proposed optimized trust technique.

Table 3
Percentage of detected nodes in DOS for 0% to 90% of compromised nodes with proposed optimized trust technique.

Percentage of compromised nodes	Percentage of detected nodes
0	100
10	95
20	93
30	93
40	88
50	84
60	86
70	82
80	81
90	77

Table 4
End to end delay in DOS for 0% to 90% of compromised nodes with proposed optimized trust technique.

Percentage of compromised nodes	Proposed trust energy aware routing	Optimized trust energy aware routing
0	0.0044	0.0041
10	0.00513	0.0047
20	0.00531	0.005
30	0.00596	0.0055
40	0.00609	0.0056
50	0.00648	0.0059
60	0.00753	0.0069
70	0.0078	0.0073
80	0.00858	0.0079
90	0.01015	0.0093

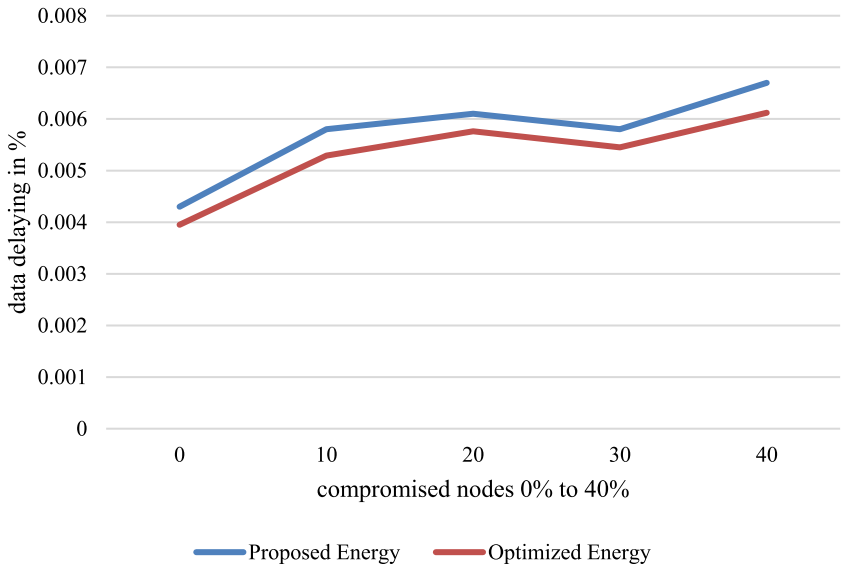


Fig. 7a. End to end delay in DDOS for 0% to 40% of compromised nodes with proposed optimized trust technique.

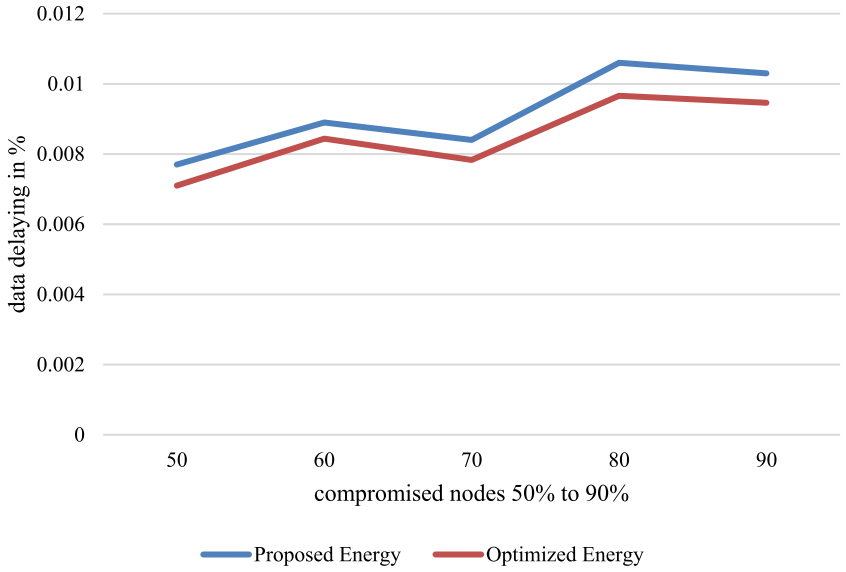


Fig. 7b. End to end delay in DDOS for 50% to 90% of compromised nodes with proposed optimized trust technique.

Table 5
Data loss rate in DDOS for 0% to 90% of compromised nodes with proposed optimized trust technique.

Percentage of compromised nodes	Proposed trust energy aware routing	Optimized trust energy aware routing
0	0.1026	0.0961
10	0.1194	0.1132
20	0.1426	0.1298
30	0.1604	0.1466
40	0.1911	0.1806
50	0.4555	0.4255
60	0.6063	0.5675
70	0.7397	0.6776
80	0.7709	0.7027
90	0.7742	0.7099

Table 6
End to end delay in DDOS for 0% to 90% of compromised nodes with proposed optimized trust technique.

Percentage of compromised nodes	Proposed trust energy aware routing	Optimized trust energy aware routing
0	0.0043	0.00395
10	0.0058	0.00529
20	0.0061	0.00576
30	0.0058	0.00545
40	0.0067	0.00612
50	0.0077	0.0071
60	0.0089	0.00844
70	0.0084	0.00783
80	0.0106	0.00966
90	0.0103	0.00946

6. Conclusion and future work

The suitability and usefulness of Cyber Physical Systems for Medical applications this review work. For the generally, cyber system vulnerabilities as secure by the CPS is challenging impacts physical systems. Unauthorized access to patients with regard to discrimination, privacy loss, abuse and physical harm. The proposed system, trust based on security to drop flooding of route request data packs in study DOS and

DDOS. Number of successful deliveries among paths used to identify malicious nodes and path data based on buffer size and neighboring nodes and available energy.

In the first phase of the work, a simple pathrater based mitigation system with an energy aware routing protocol was proposed and simulated under various malicious nodes condition. It was seen that the network collapsed as the percentage of malicious nodes increased beyond 30%. Even at 10% of malicious nodes the data loss rate was significant. The data loss rate was higher in DDOS due to cooperation of the malicious nodes.

In the second phase of the work a trust based mechanism was proposed to improve the QoS. The proposed trust based routing improved the overall routing performance even when no malicious nodes were present showing its suitability for networks without malicious nodes. In the final phase of the work optimization was used to select the optimal parameters of trust using Ant Colony metaheuristic. The results showed that the proposed trust energy aware routing increased end-to-end in DOS by 6.01% for 20 percent of compromised nodes, by 8.38% for 40 percent of compromised nodes, by 8.73% for 60 percent of compromised nodes and by 8.25% for 80 percent of compromised nodes when compared with optimized trust energy aware routing and the proposed trust energy aware routing increased end-to-end in DDOS by 5.73% for 20 percent of compromised nodes, by 9.04% for 40 percent of compromised nodes, by 5.3% for 60 percent of compromised nodes and by 9.27% for 80 percent of compromised nodes when compared with optimized trust energy aware routing. Similar improvements in end to end delay were also observed.

The present work can be extended to cloud computing side of CPS. Investigations on scheduling resources in cloud can be carried out. Integration of various entities in the CPS can be studied.

Declaration of competing interest

None.

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